

Frobenius Groups

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Introduction

The representation theory of finite groups, discovered by Frobenius towards the end of the 19th century, consists in analysing the properties and the structure of a finite group $(G, \cdot)$ by making it act linearly on a vector space. In this memoir (which was originally a TIPE, Travail d'Initiative Personnelle Encadré, for the Ecole Normale Supérieure entrance exam), we use representation theory to prove a theorem about Frobenius groups. In the appendix, we collect a few prerequisites that we will use without mention in this text.

Note that the original document was in French. It was automatically translated into English using an LLM, please signal any mistake.

1 The Frobenius theorem

Definition 1. Let $(G, \cdot)$ be a finite group with identity e and H a proper subgroup of G (i.e. $H \neq \{e\}, H \neq G$). We say that G is a Frobenius group, H a Frobenius complement, (G, H) a Frobenius pair if

$$\forall g \in G \setminus H, gHg^{-1} \cap H = \{e\}$$

Note that this is equivalent to $\forall (x, y) \in G^2, y^{-1}x \notin H \Rightarrow xHx^{-1} \cap yHy^{-1} = \{e\}$. Intuitively, this means that the conjugates of H cover G "as much as possible".

Proposition 1. If (G, H) is a Frobenius pair, $|\bigcup_{g \in G} gHg^{-1}| = |G| + 1 - \frac{|G|}{|H|}$.

Proof. $|\bigcup_{g \in G} gHg^{-1}| = |\{e\} \sqcup \bigsqcup_{gH \in G/H} (gHg^{-1} \setminus \{e\})| = 1 + |G/H|(|H| - 1)$

□

One can show that for an arbitrary subgroup H the equality above is an inequality $\leq$. It moreover shows that the conjugates of a proper subgroup never cover G (recall that G is finite, the triangular matrices provide a counter-example in $GL_n(\mathbb{C})$ which is infinite).

Example 1. Let G be a finite group acting transitively on a set X , such that for $g \in G \setminus \{e\}$, g acts fixing at most one point of X . Then if $\alpha \in X$ such that $\text{Stab}(\alpha) \neq \{e\}$, $(G, \text{Stab}(\alpha))$ is a Frobenius pair.

Proof. We have $\text{Stab}(\alpha) \neq G$ (by transitivity). If $g \in G \setminus \text{Stab}(\alpha)$, then $g \cdot \alpha \neq \alpha$ so $g \cdot \alpha$ and α are not fixed at the same time other than by e , hence $\text{Stab}(g \cdot \alpha) \cap \text{Stab}(\alpha) = \{e\}$ that is $g\text{Stab}(\alpha)g^{-1} \cap \text{Stab}(\alpha) = \{e\}$. □

One can in fact show that every Frobenius group is of this form. Here are some examples of them:

Example 2. • If $\mathbb{K}$ is a finite field and $G = \text{Aff}(1, \mathbb{K}) = \{\tau_{a,b} : x \in \mathbb{K} \mapsto ax + b, (a, b) \in \mathbb{K}^* \times \mathbb{K}\}$ is the group of affine bijections of $\mathbb{K}$ onto itself and $H(\mathbb{K}) = \{\tau_{a,0} : x \in \mathbb{K} \mapsto ax, a \in \mathbb{K}^*\}$ those that fix 0, then (G, H) is a Frobenius pair ($\tau_{a,b}$ fixes exactly $\frac{b}{1-a}$ for $a \neq 1$, fixes no point of $\mathbb{K}$ for $a = 1$ and $b \neq 0$, fixes all of $\mathbb{K}$ for $a = 1$ and $b = 0$ that is for $\tau_{a,b} = \text{id}$; finally $H(\mathbb{K}) = \text{Stab}(0)$)

- If $G = \mathcal{A}_4$ (alternating group) and γ a 3-cycle, then $(G, \langle \gamma \rangle)$ is a Frobenius pair
- If $n \in \mathbb{N}^*$ is odd, D_n the dihedral group (affine transformations of the plane preserving the regular n -gon centred at the origin), s an orthogonal symmetry in D_n , then $(D_n, \langle s \rangle)$ is a Frobenius pair.

Theorem 1 (Frobenius). *Let (G, H) be a Frobenius pair. We set*

$$N = \left(G \setminus \bigcup_{g \in G} gHg^{-1} \right) \cup \{e\}$$

called the Frobenius kernel. Then N is a normal subgroup of G and $N \cap H = \{e\}$ and $G = NH$ (we say that G is the semidirect product of N by H).

$N \cap H = \{e\}$ is obvious, moreover thanks to proposition 1, $|N| = |G| - |\bigcup_{g \in G} gHg^{-1}| + 1 = |G|/|H|$. Granting that N is a subgroup of G , $\varphi : (n, h) \in N \times H \mapsto nh$ is injective (if $nh = n'h'$ then $n'^{-1}n = h'h^{-1} \in N \cap H = \{e\}$ so $(n, h) = (n', h')$) then, by cardinality, surjective hence $G = NH$. Moreover if $g \in G$, $gNg^{-1} \subseteq N$ (otherwise an element of $N \setminus \{e\}$ would be conjugate to an element of H contradicting the definition of N) so N is normal. The difficult part of the theorem lies in the fact that N is a subgroup. This is what we will prove in section 5.

Example 3. We illustrate this with the pairs of example 2. In the "group action" formulation of example 1, the Frobenius kernel is the set of elements of G acting without a fixed point, to which we add the identity e . Thus,

- The kernel of $(\mathcal{A}_4, < \gamma >)$ is the Klein group $V = \{id, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$
- The kernel of $(\text{Aff}(1, \mathbb{K}), H(\mathbb{K}))$ is the subgroup $\{\tau_{1,b}, b \in \mathbb{K}\}$ of translations of $\text{Aff}(1, \mathbb{K})$
- The Frobenius kernel of D_n is the group of rotations centered at the origin by an angle in $\frac{2\pi}{n}\mathbb{Z}$ (naturally isomorphic to $\mathbb{U}_n$)

2 Representations of a finite group

To prove the Frobenius theorem we will seek to express N as the kernel of a representation.

Definition 2. *Let $(G, \cdot)$ be a finite group. We call a representation of G a pair (ρ, V) where V is a finite-dimensional $\mathbb{C}$ -vector space, and ρ a morphism from G to $\text{GL}(V)$. We will sometimes write only ρ . We call the degree of ρ the integer $\dim(V)$.*

We thus view the elements of G as linear transformations on V .

Example 4. • Representations of degree 1 are identified with the morphisms from G to $\mathbb{C}^*$. In particular we call the trivial representation the one corresponding to the trivial morphism from G to $\mathbb{C}^*$.

- A more elaborate example: if $(f_g)_{g \in G}$ is a basis indexed by G of a vector space V of dimension $|G|$, by setting for $g \in G$, $\rho(g)$ the unique element of $\text{GL}(V)$ satisfying $\forall h \in G, \rho(g)(f_h) = f_{gh}$, we define a representation of G called "regular". We moreover say that it is "faithful" because $\ker(\rho) = \{e\}$.
- If (ρ, V) and (ρ', V') are 2 representations of G , then by setting for $g \in G$, we define a representation $((\rho, \rho'), \mathcal{L}(V, V'))$ by setting: $\forall u \in \mathcal{L}(V, V'), (\rho, \rho')(g)(u) = \rho'(g) \circ u \circ \rho(g^{-1})$,
- Let (ρ, V) and (ρ', V') be as above. Then we define a representation $(\rho \oplus \rho', V \times V')$ by setting $\forall (g, v, v') \in G \times V \times V', (\rho \oplus \rho')(g)((v, v')) = (\rho(g)(v), \rho'(g)(v'))$. This construction also works for subspaces in direct sum on which ρ and ρ' are defined: in a basis adapted to this decomposition, the matrices of the $(\rho \oplus \rho')(g)$ will consist of 2 square diagonal blocks of sizes $\dim(V)$ and $\dim(V')$.

Definition 3. *If (ρ, V) and (ρ', V') are 2 representations of G , we say that ρ is equivalent to ρ' , and we write $\rho \sim \rho'$ if $\dim(V) = \dim(V')$ and if there exist bases e, e' of V and V' such that*

$$\forall g \in G, \text{Mat}_e(\rho(g)) = \text{Mat}_{e'}(\rho'(g))$$

i.e. if there exists $u \in \mathcal{L}(V, V')$ an isomorphism such that $\forall g \in G, u \circ \rho(g) = \rho'(g) \circ u$. Moreover for arbitrary ρ, ρ' , we write $\text{Hom}(\rho, \rho') = \{u \in \mathcal{L}(V, V'), \forall g \in G, u \circ \rho(g) = \rho'(g) \circ u\}$ which is clearly a subspace of $\mathcal{L}(V, V')$.

Definition 4. *A representation ρ of G is said to be irreducible if the only subspaces stable under all the $\rho(g)$ for $g \in G$ are $\{0\}$ and V .*

These representations cannot be "broken" into subrepresentations, this is the case for example of the representations of degree 1. Indeed, we have the following proposition:

Proposition 2 (Maschke). *Let (ρ, V) be a representation of G and U a subspace of V stable under all the $\rho(g)$ for $g \in G$. There exists a complement S of U which is itself stable under all the $\rho(g)$.*

Proof. It suffices to construct a projector with image U which commutes with all the $\rho(g)$, its kernel will then be a desired complement. Let p be an arbitrary projector with image U . Let us modify it by averaging over G :

$$q = \frac{1}{|G|} \sum_{g \in G} \rho(g) \circ p \circ \rho(g^{-1})$$

It is indeed an endomorphism with image included in U such that $q|_U = id_U$ (because the $\rho(g)$ stabilize U) and which commutes with the $\rho(g)$ ($\rho(g) \circ q \circ \rho(g)^{-1} = q$ because $h \in G \mapsto gh \in G$ is bijective).

□

By induction on the dimension of V we thus show the following co-reduction result:

Proposition 3. *If (ρ, V) is a representation, there exist $k \in \mathbb{N}^*$ and $V_1, V_2, \dots, V_k$, subspaces of V , such that $V = \bigoplus_{i=1}^k V_i$, and irreducible representations $(\rho_1, V_1), \dots, (\rho_k, V_k)$ such that*

$$\rho = \rho_1 \oplus \rho_2 \oplus \dots \oplus \rho_k$$

Finally, here is a last proposition which will prove very useful:

Proposition 4 (Schur). *Let $(\rho, V), (\rho', V')$ be 2 irreducible representations of G . Then $\dim \text{Hom}(\rho, \rho') = \mathbb{1}_{\rho \sim \rho'}$*

Proof. • Suppose $\rho \sim \rho'$. Up to transporting ρ' by the isomorphism given by the equivalence, suppose $V = V'$ and $\rho = \rho'$. Let $u \in \text{Hom}(\rho, \rho)$. Since $\mathbb{C}$ is algebraically closed, u has an eigenvalue λ . Then for all g in G , u and $\rho(g)$ commute so $E_\lambda(u) = \ker(u - \lambda \text{id})$ is stable under $\rho(g)$. By irreducibility of ρ , since $\ker(u - \lambda \text{id})$ is nonzero, it is V and $u = \lambda \text{id}$. Conversely, the $\lambda \text{id}, \lambda \in \mathbb{C}$ are in $\text{Hom}(\rho, \rho)$. Finally $\text{Hom}(\rho, \rho) = \mathbb{C} \text{id}$ is of dimension 1.

- Suppose $\rho \not\sim \rho'$. Let $u \in \text{Hom}(\rho, \rho')$. If $g \in G$, $\rho'(g) \circ u = u \circ \rho(g)$ so $\text{im}(u)$ is stable under $\rho'(g)$. By irreducibility, $\text{im}(u) = V'$ or $\{0\}$. Likewise, $\ker(u) = V$ or $\{0\}$. So since u is not an isomorphism, $\ker(u) = V$ and $\text{im}(u) = \{0\}$ and $u = 0$ which concludes.

□

3 Characters

A good portion of the information contained in a representation is in fact stored in its character:

Definition 5. *Let (ρ, V) be a representation of G . We call the character of ρ the function*

$$\begin{aligned} \chi_\rho : G &\longrightarrow \mathbb{C} \\ g &\longmapsto \text{Tr}(\rho(g)) \end{aligned}$$

We call a character any function in $\mathbb{C}^G$ which is the character of a representation. We say that it is irreducible if it comes from an irreducible representation

Example 5. • $\chi(e) = \text{Tr}(\text{id})$ is the degree of the representation

- 2 equivalent representations have the same character (same matrices in well-chosen bases)
- If $g, h \in G$, χ a character, $\chi(ghg^{-1}) = \text{Tr}(\rho(g)\rho(h)\rho(g)^{-1}) = \text{Tr}(\rho(h)) = \chi(h)$. We denote by $\text{Cent}(G)$ the subspace of central functions of $\mathbb{C}^G$, i.e. constant on the conjugacy classes of G (of dimension the number of such classes). Then $\chi \in \text{Cent}(G)$.
- We trivially have $\chi_{\rho_1 \oplus \rho_2} = \chi_{\rho_1} + \chi_{\rho_2}$
- For $g \in G$, $g^{|G|} = e$ so $X^{|G|} - 1 = \prod_{\omega \in \mathbb{U}_{|G|}} (X - \omega)$ annihilates $\rho(g)$ and is split with simple roots so $\rho(g)$ is diagonalizable with spectrum in $\mathbb{U}_{|G|}$. If $\{\lambda_1, \dots, \lambda_d\}$ is the spectrum of $\rho(g)$, $\{\lambda_1^{-1}, \dots, \lambda_d^{-1}\}$ is that of $\rho(g)^{-1} = \rho(g^{-1})$ hence, since the spectrum is in $\mathbb{U}$, $\chi(g^{-1}) = \lambda_1^{-1} + \dots + \lambda_d^{-1} = \overline{\lambda_1 + \dots + \lambda_d} = \overline{\chi(g)}$
- Using the notations of example 4, $\chi_{\widetilde{(\rho, \rho')}} = \overline{\chi_\rho} \chi_{\rho'}$ (a consequence, using the previous point and the definition of $\widetilde{(\rho, \rho')}$, of the fact that for $A, B \in \mathcal{M}_n(\mathbb{C})$, the linear map $M \in \mathcal{M}_n(\mathbb{C}) \mapsto AMB \in \mathcal{M}_n(\mathbb{C})$ has trace $\text{Tr}(A)\text{Tr}(B)$, because if $1 \leq i, j \leq n$, $AE_{i,j}B = \sum_{1 \leq k, l \leq n} A_{ki}B_{jl}E_{k,l}$ has coordinate $A_{ki}B_{jj}$ at $E_{i,j}$)

Definition/Proposition 1. *We endow $\mathbb{C}^G$ with the Hermitian inner product (cf. appendix) defined by:*

$$\forall (u, v) \in \mathbb{C}^G \times \mathbb{C}^G, \langle u, v \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{u(g)} v(g)$$

To compute the Hermitian product of 2 characters, we again use the "averaging technique":

Lemma 1. *Let ρ be a representation, then $\frac{1}{|G|} \sum_{g \in G} \rho(g)$ is a projector onto $V^G := \bigcap_{g \in G} \ker(\rho(g) - \text{id})$*

Proof. If $x \in V^G$, $\frac{1}{|G|} \sum_{g \in G} \rho(g)(x) = \frac{1}{|G|} \sum_{g \in G} x = x$ (1). Moreover, if $x \in V$ and $y = \frac{1}{|G|} \sum_{g \in G} \rho(g)(x)$, and $h \in G$, $\rho(h)(y) = \frac{1}{|G|} \sum_{g \in G} \rho(hg)(x) = y$ (because $g \mapsto hg$ is bijective) and so $y \in V^G$ which concludes (2). With (1) and (2), we obtain the desired result. □

Proposition 5. *If $(\rho, V), (\rho', V')$ are 2 representations of G , $\langle \chi_\rho, \chi_{\rho'} \rangle = \dim(\text{Hom}(\rho, \rho'))$*

Proof. We set $\widetilde{(\rho, \rho')}$ as in example 4. We apply lemma 1 to this representation. On the one hand, $\bigcap_{g \in G} \ker(\widetilde{(\rho, \rho')}(g) - \text{id}_{\mathcal{L}(V, V')}) = \text{Hom}(\rho, \rho')$, so $\text{Tr}(\frac{1}{|G|} \sum_{g \in G} \widetilde{(\rho, \rho')}(g)) = \dim(\text{Hom}(\rho, \rho'))$ (by diagonalizing a projector, we see that its trace is its rank). On the other hand, by linearity and example 5, $\text{Tr}(\frac{1}{|G|} \sum_{g \in G} \widetilde{(\rho, \rho')}(g)) = \frac{1}{|G|} \sum_{g \in G} \chi_{\widetilde{(\rho, \rho')}}(g) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_\rho(g)} \chi_{\rho'}(g) = \langle \chi_\rho, \chi_{\rho'} \rangle$ □

By combining with Schur's lemma (proposition 4),

Corollary 1. *If $(\rho, V), (\rho', V')$ are 2 irreducible representations of G , $\langle \chi_\rho, \chi_{\rho'} \rangle = \mathbb{1}_{\rho \sim \rho'}$. Thus, the family of irreducible characters is orthonormal hence finite ($\mathbb{C}^G$ is finite-dimensional). Moreover, 2 non-equivalent irreducible representations have distinct characters: there are therefore as many irreducible representations up to equivalence as there are irreducible characters.*

Writing $\rho_1, \dots, \rho_r$ a fundamental system of irreducible representations, and $\chi_1, \dots, \chi_r$ the associated characters, by combining with proposition 3,

Corollary 2. *If ρ is a representation, there exists $(n_1, \dots, n_r) \in \mathbb{N}^r$ such that $\rho \sim n_1 \rho_1 \oplus \dots \oplus n_r \rho_r$ (where $n_i \rho_i$ denotes $\underbrace{\rho_i \oplus \dots \oplus \rho_i}_{n_i \text{ times}}$). Then $\chi = \sum_{i=1}^r n_i \chi_i$*

Proposition 6. *If ρ and ρ' are 2 representations, $\rho \sim \rho' \iff \chi_\rho = \chi_{\rho'}$*

Proof. The direct implication is clear. If $\chi_\rho = \chi_{\rho'} =: \chi$, by writing $\rho \sim n_1 \rho_1 \oplus \dots \oplus n_r \rho_r$ and $\rho' \sim n'_1 \rho_1 \oplus \dots \oplus n'_r \rho_r$, by orthonormality $\langle \chi_i, \chi_\rho \rangle = \sum_{j=1}^r n_j \langle \chi_i, \chi_j \rangle = n_i$. Likewise, $\langle \chi_i, \chi_{\rho'} \rangle = n'_i$. Then $n_i = n'_i$ and $\rho \sim \rho'$. $\square$

We can in fact show better than the orthonormality of the characters (we will not prove it here):

Proposition 7. *The irreducible characters $\chi_1, \dots, \chi_r$ form an orthonormal basis of $\text{Cent}(G)$. Thus r is the number of conjugacy classes in G and if $f \in \text{Cent}(G)$,*

$$f = \sum_{i=1}^r \langle \chi_i, f \rangle \chi_i$$

4 Induced representations

Let H be a subgroup of G . From a representation (ρ, V) of G , we can easily obtain the representation $(\rho|_H, V)$ of H , denoted $\rho|_H$. Conversely, the aim of this section is to give a natural way to extend a representation (ρ_0, W) of H into a representation (ρ, V) of G (where $W \subseteq V$). We write

$$G = \bigsqcup_{i=1}^l g_i H$$

the partition of G into left cosets modulo H (where the g_i are choices of representatives and $g_1 H = H$).

Definition 6. *Let (ρ, V) be a representation of G , and W a subspace of V stable under the $\rho(h), h \in H$. Let $\rho_0 = \rho|_{H,W}$ be the restriction of ρ to H and to the subspace W . We say that ρ is induced from ρ_0 and we write $\rho = \text{Ind}_H^G(\rho_0)$ if*

$$V = \bigoplus_{i=1}^l \rho(g_i)(W) \quad (*)$$

Note that under the hypotheses of definition 6, if $g \in G$ is written $g_i h$ where $h \in H$, $\rho(g)(W) = \rho(g_i h)(W) = \rho(g_i) \circ \rho(h)(W) = \rho(g_i)(W)$. So the $\rho(g_i)(W)$ do not depend on the choice of representatives in the left coset modulo H . We have the following proposition:

Proposition 8. *If ρ' is a representation of H , there exists ρ (unique up to equivalence) a representation of G such that $\rho = \text{Ind}_H^G(\rho')$*

We do not prove it: let us merely observe that if (ρ_0, W) and the action of the $\rho(g_i)$ on W are fixed, the entire induced representation is. Indeed, it suffices that it be so on the $\rho(g_i)(W)$ (by direct sum) and if $g \in G$, $1 \leq i \leq r$, $w \in W$, we have $\rho(g)(\rho(g_i)(w)) = \rho(g_j)(\rho_0(h)(w))$ where $gg_i = g_j h$ (we look at the representative of the class of gg_i modulo H). The following result justifies the relevance of induced representations:

Proposition 9. *Let (ρ, V) and (ρ', V') be 2 representations of G and (ρ_0, W) a representation of H . We suppose $\rho = \text{Ind}_H^G(\rho_0)$ (in particular, $\rho_0 = \rho|_{H,W}$). Then $\text{Hom}(\rho, \rho') \simeq \text{Hom}(\rho_0, \rho'_|_H)$*

Proof. $u \in \text{Hom}(\rho, \rho') \subseteq \mathcal{L}(V, V') \mapsto u|_W \in \text{Hom}(\rho_0, \rho'_|_H) \subseteq \mathcal{L}(W, V')$ is linear. For bijectivity, let us show that an element $v \in \text{Hom}(\rho_0, \rho'_|_H)$ extends on W into a unique element of $\text{Hom}(\rho, \rho')$: if u is such an extension, $\forall 1 \leq i \leq l, \forall w \in W, u(\rho(g_i)(w)) = \rho'(g_i)(u(w)) = \rho'(g_i)(v(w))$ which defines u on the whole of V by the direct sum (*). Conversely, one checks without difficulty that such a u works. $\square$

By translating proposition 9 into an equality of dimensions (thanks to proposition 5),

Corollary 3. *Using the notations of proposition 9, by writing $\text{Ind}_H^G(\chi_{\rho_0}) = \chi_\rho$, and $\langle \cdot, \cdot \rangle_G$ (resp. $\langle \cdot, \cdot \rangle_H$) the canonical inner product on $\text{Cent}(G)$ (resp. $\text{Cent}(H)$), we have*

$$\langle \text{Ind}_H^G(\chi_{\rho_0}), \chi_{\rho'} \rangle_G = \langle \chi_{\rho_0}, (\chi'_{\rho'})|_H \rangle_H$$

Thanks to proposition 6, we can speak of the "induced character" of χ_{ρ_0} for $\text{Ind}_H^G(\chi_{\rho_0})$.

Proposition 10. *If χ is a character of H ,*

$$\forall g \in G, \text{Ind}_H^G(\chi)(g) = \frac{1}{|H|} \sum_{k \in G, kgk^{-1} \in H} \chi(kgk^{-1})$$

Proof. We write (ρ, V) and (ρ', W) such that $\chi = \chi_{\rho'}$ and $\rho = \text{Ind}_H^G(\rho')$. Let $g \in G$. $\rho(g)$ permutes the $\rho(g_i)(W)$ hence, by looking at the matrices of the $\rho(g)$ in a basis adapted to $V = \bigoplus_{i=1}^l \rho(g_i)(W)$, to compute the trace of $\rho(g)$, it suffices to consider those i such that $\rho(g)(\rho(g_i)(W)) = \rho(g_i)(W)$ i.e. such that $\rho(g_i^{-1}gg_i)(W) = W$ i.e. $g_i^{-1}gg_i \in H$. Thus, by invariance of the trace under change of basis (we transport $\rho(g_i)(W)$ onto W by the isomorphism $\rho(g_i)$) $\chi(g) = \sum_{1 \leq i \leq l, g_i^{-1}gg_i \in H} \text{Tr}(\rho(g)|_{\rho(g_i)(W)}) = \sum_{1 \leq i \leq l, g_i^{-1}gg_i \in H} \text{Tr}(\rho(g_i^{-1}gg_i)|_W) = \sum_{1 \leq i \leq l, g_i^{-1}gg_i \in H} \chi(g_i^{-1}gg_i)$ which concludes because χ is central on H and the right cosets modulo H have cardinality $|H|$. $\square$

This formula allows us to define $\text{Ind}_H^G(f)$ for any central function f (we replace χ by f in the formula). Since the characters linearly span the central functions, we deduce from corollary 3:

Proposition 11. *If $f_1 \in \text{Cent}(H)$ and $f_2 \in \text{Cent}(G)$, we have*

$$\langle \text{Ind}_H^G(f_1), f_2 \rangle_G = \langle f_1, f_2|_H \rangle_H$$

5 Proof of the Frobenius theorem

Time for the proof of theorem 1. We thus fix a Frobenius pair (G, H) . We denote by N the corresponding Frobenius kernel. The central result of the proof is the following proposition:

Proposition 12. *Let χ be an irreducible character of H . There exists $\tilde{\chi}$ an irreducible character of G such that $\tilde{\chi}|_H = \chi$ and $\tilde{\chi}$ is constant on N .*

Remark 1. *One can easily see that every central function on H extends in a unique way into a central function on G constant on N . What matters here is that this extension is a character (irreducible).*

Here is a lemma which will allow us to produce characters constant on N :

Lemma 2. *Let $f \in \text{Cent}(H)$ such that $f(e) = 0$ and $\tilde{f} = \text{Ind}_H^G(f)$. Then $\tilde{f}|_H = f$ and $\tilde{f}|_N = 0$*

Proof. It is obvious by the formula defining the induced function that $\tilde{f}(e) = 0 = f(e)$.

Let $g \in H \setminus \{e\}$. Notice that if $k \in G$, $kgk^{-1} \in H \iff g \in k^{-1}Hk \cap H \iff k \in H$ by definition of a Frobenius complement. We thus have $\tilde{f}(g) = \frac{1}{|H|} \sum_{k \in G, kgk^{-1} \in H} f(kgk^{-1}) = \frac{1}{|H|} \sum_{k \in H} f(kgk^{-1}) = \frac{1}{|H|} \cdot |H|f(g) = f(g)$ because f is central on H .

Let $g \in N \setminus \{e\}$, for $k \in G$, $kgk^{-1} \notin H$ (otherwise, $g \in k^{-1}Hk$, absurd by definition of N), hence $\tilde{f}(g) = 0$. $\square$

Proof of proposition 12. One may refer to the table in remark 2 after the proof which clarifies the functions in play. We reuse the notations of proposition 12.

Let $d = \chi(e)$ be the degree of χ , $l = |G|/|H| \in \mathbb{N}$, θ_H the trivial character of H and θ_G that of G . If χ is the trivial character, it suffices to set $\tilde{\chi}$ the trivial character of G which is irreducible.

Otherwise, to apply lemma 2, we consider $\chi - d \cdot \theta_H \in \text{Cent}(H)$ which vanishes at e .

Then, $\tilde{\psi} := \text{Ind}_H^G(\chi - d \cdot \theta_H)$ vanishes on N and admits $\chi - d \cdot \theta_H$ as a restriction to H (*). However it is not a character (because $\tilde{\psi}(e) = 0$). In order to obtain an irreducible character, we would need at the very least a central function of Hermitian norm 1. Let us therefore compute, with proposition 11,

$$\langle \tilde{\psi}, \tilde{\psi} \rangle_G = \langle \text{Ind}_H^G(\chi - d \cdot \theta_H), \tilde{\psi} \rangle_G = \langle \chi - d \cdot \theta_H, \tilde{\psi}|_H \rangle_H = \langle \chi - d \cdot \theta_H, \chi - d \cdot \theta_H \rangle_H = 1 + d^2$$

where we have used the orthogonality of the irreducible characters and the fact that $\chi \neq \theta_H$. According to proposition 11,

$$\langle \tilde{\psi}, \theta_G \rangle_G = \langle \text{Ind}_H^G(\chi - d \cdot \theta_H), \theta_G \rangle_G = \langle \chi - d \cdot \theta_H, \theta_G|_H \rangle_H = \langle \chi - d \cdot \theta_H, \theta_H \rangle_H = -d$$

We are thus led to set $\tilde{\chi} = \tilde{\psi} + d \cdot \theta_G$ which yields (where the inner products are taken in $\text{Cent}(G)$)

$$\langle \tilde{\chi}, \tilde{\chi} \rangle = \langle \tilde{\psi}, \tilde{\psi} \rangle + 2d\langle \tilde{\psi}, \theta_G \rangle + d^2\langle \theta_G, \theta_G \rangle = 1 + d^2 - 2d^2 + d^2 = 1$$

Moreover, $\tilde{\chi}$ is manifestly a linear combination of characters with coefficients in $\mathbb{Z}$. There therefore exist $n_1, \dots, n_r \in \mathbb{Z}$ such that $\tilde{\chi} = \sum_{i=1}^r n_i \chi_i$. Then $1 = \langle \tilde{\chi}, \tilde{\chi} \rangle = n_1^2 + \dots + n_r^2$ hence since they are in $\mathbb{Z}$, only one of the n_i is nonzero and equals ± 1 . Since $\tilde{\chi}(e) = d > 0$ (cf. below), this n_i equals 1 and $\tilde{\chi}$ is an irreducible character.

Finally, $\tilde{\chi}$ remains constant on N (we added $d \cdot \theta_G$ to $\tilde{\psi}$ which was constant on N), and for $h \in H$, $\tilde{\chi}(h) = \tilde{\psi}(h) + d \cdot \theta_G(h) = (\chi - d \cdot \theta_H)(h) + d = \chi(h)$ then $\tilde{\chi}|_H = \chi$, which concludes. $\square$

Remark 2. We in fact know all the values taken by $\tilde{\chi}$. Indeed, $G = N \cup \bigcup_{g \in G} gHg^{-1}$, and no element of N is conjugate to an element of $\bigcup_{g \in G} gHg^{-1} \setminus \{e\}$ (otherwise it would be conjugate to an element of $H \setminus \{e\}$). Moreover, every element of $\bigcup_{g \in G} gHg^{-1} \setminus \{e\}$ is conjugate to an element of H . Hence if we write $A_1, \dots, A_m$ the conjugacy classes in G of the elements of $\bigcup_{g \in G} gHg^{-1} \setminus \{e\}$, we have $\bigcup_{g \in G} gHg^{-1} \setminus \{e\} = \sqcup_{i=1}^m A_i$ and we have $h_1 \in A_1 \cap H, \dots, h_m \in A_m \cap H$. We then have the following table (since the characters are central)

	e	A_1	$\dots$	A_m	$N \setminus \{e\}$
$\tilde{\psi}$	0	$\chi(h_1) - d$	$\dots$	$\chi(h_m) - d$	0
θ_G	1	1	$\dots$	1	1
$\tilde{\chi} = \tilde{\psi} + d \cdot \theta_G$	$d = \chi(e)$	$\chi(h_1)$	$\dots$	$\chi(h_m)$	$d = \chi(e) = \tilde{\chi}(e)$

The decomposition of a character into irreducible characters in fact yields the corollary:

Corollary 4. Let χ be a character of H . There exists $\tilde{\chi}$ a character of G such that $\tilde{\chi}|_H = \chi$ and $\tilde{\chi}$ is constant on N . The table above remains valid.

A last lemma will allow us to conclude.

Lemma 3. Let χ be a character of G coming from a representation ρ . Then $\chi(g) = \chi(e)$ iff $g \in \ker(\rho)$

Proof. Let $d = \chi(e)$ be the degree of ρ . If $g \in G$, as seen in example 5, $\rho(g)$ is diagonalizable with spectrum $\{\lambda_1, \dots, \lambda_d\}$ in $\mathbb{U}$. Then the triangle inequality gives $|\chi(g)| = |\lambda_1 + \dots + \lambda_d| \leq |\lambda_1| + \dots + |\lambda_d| = d = \chi(e)$ with equality iff $\lambda_1, \dots, \lambda_d$ are positively collinear hence iff they are equal (because of modulus 1). So $\lambda_1 + \dots + \lambda_d = d$ iff $\lambda_1 = \dots = \lambda_d = 1$ iff $\rho(g) = id$. $\square$

At this stage, all the ingredients are in place to conclude.

End of the proof of theorem 1. We take ρ a faithful representation of H (for example the regular representation) and χ its character. Let $\tilde{\chi}$ be as defined in corollary 4 and $\tilde{\rho}$ the associated representation. Let us show that

$$N = \ker(\tilde{\rho})$$

We have, according to the table of remark 2 and lemma 3, $\ker(\tilde{\rho}) = \{x \in G, \tilde{\chi}(x) = \tilde{\chi}(e)\} = N$ (where for the last equality we used the following fact: $\chi(h_1), \dots, \chi(h_m) \neq \chi(e)$ according to lemma 3 and the faithfulness of ρ).

N is therefore a (normal) subgroup of G , which concludes. $\square$

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6 Appendix: useful prerequisites

Throughout this part, let $(G, \cdot)$ be a finite group.

Definition A. Let X be a set. We call an action of the group G on X any morphism φ from G to $\mathfrak{S}(X)$. We write $g \cdot x := \varphi(g)(x)$ for $(g, x) \in G \times X$. If $x \in X$, we call the stabilizer of x the set $\text{Stab}(x) = \{g \in G, g \cdot x = x\}$. One checks that it is a subgroup of G . We say that the action is faithful if φ is injective. We say that the action is transitive if $\forall x \in X, \{g \cdot x, g \in G\} = X$.

Proposition A. Suppose that G acts on X . Let $g \in G$ and $x \in X$. $\text{Stab}(g \cdot x) = g\text{Stab}(x)g^{-1}$.

Proof. If $h \in H$, $h \in \text{Stab}(g \cdot x) \iff h \cdot (g \cdot x) = g \cdot x \iff (g^{-1}hg) \cdot x = x \iff g^{-1}hg \in \text{Stab}(x) \iff h \in g\text{Stab}(x)g^{-1}$ $\square$

Definition B. Let $(G, \cdot)$ be a group.

- If $g \in G$, the map $\varphi_g : x \in G \mapsto gxg^{-1} \in G$ is a group automorphism, called the conjugation automorphism, and the map $g \in G \mapsto \varphi_g \in \text{Aut}(G)$ is an action of G on itself.
- x and y in G are said to be conjugate if there exists $g \in G$ such that $x = \varphi_g(y)$ which defines an equivalence relation on G whose classes are called conjugacy classes. We define likewise the conjugates of a subset of G . If H is a subgroup of G , for every $g \in G$, $\varphi_g(H)$ is a subgroup of G (as the image of a subgroup under a morphism) of cardinality $|H|$ (because φ_g is injective).

Definition C. Let H be a subgroup of G . We say that H is "distinguished" or "normal" if

$$\forall g \in G \forall h \in H, ghg^{-1} \in H$$

Definition D. Let H be a subgroup of G . We say that $x \in G$ and $y \in G$ are "congruent modulo H " if $y^{-1}x \in H$. One checks that this defines an equivalence relation on G whose classes ("left cosets modulo H ") are the $gH, g \in G$. We use the notation $G/H = \{gH, g \in G\}$. Since $\tau_g : h \in G \mapsto gh$ is injective for $g \in G$, the elements of G/H all have cardinality $|H|$ hence, since their union is G , $|G/H| = |G|/|H|$ (which shows in particular that $|H|$ divides $|G|$: this is Lagrange's theorem).

Finally, we define the notion of Hermitian space which generalizes that of Euclidean space:

Definition E. Let V be a $\mathbb{C}$ -vector space. We call a Hermitian inner product on V any map $\langle, \rangle : V^2 \rightarrow \mathbb{C}$ which is linear in the 2nd variable ($\forall (x, y, z, \lambda) \in V^3 \times \mathbb{C}, \langle x, y + \lambda z \rangle = \langle x, y \rangle + \lambda \langle x, z \rangle$), Hermitian symmetric ($\forall (x, y) \in V^2, \langle x, y \rangle = \overline{\langle y, x \rangle}$), positive ($\forall x \in V, \langle x, x \rangle \in \mathbb{R}^+$), definite ($\forall x \in V, \langle x, x \rangle = 0 \iff x = 0$). We define as in the Euclidean case the notion of orthogonality and of orthonormal families.